

Onboarding

Julia on HLRS Cluster/Laptops

HLRS Laptops

Most of the course can be completed on the laptops.

Equipped with NVIDIA GPU

Jupyter + VS Code

jupyter lab
code

Course materials

[\\$HOME/JuliaHLRS25](#)

Julia VS Code integration via extension.

The screenshot displays the VS Code interface with the Julia extension. The Explorer sidebar on the left shows the project structure, including folders like `.github`, `.vscode`, `benchmark`, `bin`, `deps`, `docs`, `src`, `test`, `.gitignore`, `.travis.yml`, `appveyor.yml`, and `LICENSE.md`. The Julia Explorer sidebar shows the workspace structure, including folders for `Core`, `InteractiveUtils`, `PlotThemes`, `Plots`, and a `Julia REPL`. The main editor shows a Julia script with the following code:

```
9      ...end
10      ...return maxiter
11  end mandel (generic function with 1 method)
12
13  for i in 1:10
14      ...println(i)
15  end
16
17  map
```

A tooltip for the `map` function is visible, providing the following information:

- `map(f, c...) -> collection`
- Transform collection `c` by applying `f` to each element. For multiple collection arguments, apply `f` elementwise.
- See also: `mapslices`
- Examples**
- `julia> map(x -> x * 2, [1, 2, 3])`
- `3-element Array{Int64,1}:`
- `2`

The bottom panel shows the Julia REPL with the output of the script:

```
4
5
6
7
8
9
10
```

The status bar at the bottom indicates the current file is `sp/isdeflocals*`, the environment is `Julia env: v1`, and the cursor is at `Ln 17, Col 4` with `Spaces: 4` and `UTF-8 LF` encoding.

Let's get started!

→ `cd JuliaHLRS25`

→ `git pull`

→ `jupyter lab`

→ `exercises/Day1/1_onboarding/laptop.md`

HLRS Training Cluster

The cluster has two types of nodes.

CPU nodes

“skl”

2 CPUs (skylake)

40 cores total

GPU nodes

“clx-ai”

2 CPUs (cascade lake)

36 cores total

8x NVIDIA V100

Jobs are scheduled with PBS Pro.

Submit a job:

```
qsub job_script.sh
```

Check on your queued/running jobs:

```
qstat -nw
```


VS Code → HLRS Cluster

VS Code → Cluster via SSH

Login node

Works fine, just connect to

`accountname@training.hlr.de`

Compute node

At HLRS, possible but inconvenient

`SetEnv PBS_JOBID=...`

`SSH ProxyJump`

To get Julia, load the necessary system modules.

Modules on the HLRS training cluster

```
module use juliahpc  
module use mpi/openmpi  
# shortcut: ml juliahpc
```

Outside of the course: If there is no (working) system module, use standard binaries provided by [juliaup](#).

Comment: Julia depot is on the parallel file system.

`JULIA_DEPOT_PATH` = where Julia stores stuff
packages
binary dependencies
...

Why not `$HOME`?

Quotas

Can be read-only for compute jobs

Comment: Julia VSCode extension requires a wrapper.

```
[...]
```

```
# Load modules
```

```
module load juliahpc
```

```
module load mpi/openmpi
```

```
# Pass on all arguments
```

```
exec julia "${@}"
```

Let's try it!

→ [exercises/Day1/1_onboarding/cluster.md](#)

